SCGNet: Semantic Correlation-Guided Network for Remote Sensing Semantic Change Detection

Xu Tang, *Senior Member, IEEE*, Weihao Jiang, Jingjing Ma, *Member, IEEE*, Xiangrong Zhang, *Senior Member, IEEE*, Fang Liu, *Member, IEEE*, and Licheng Jiao, *Life Fellow, IEEE*

***Abstract*—Semantic change detection (SCD) is an important research task in remote sensing (RS), as it not only identifies changed regions but also characterizes "from–to" semantic transitions, which are of great value in practical applications. However, under multitask learning architectures, task-coupling interference between semantic segmentation (SS) and binary change detection, together with the pronounced intraclass variance and subtle interclass differences in bitemporal RS images, poses significant challenges to SCD. This article proposes a new SCD method called the semantic correlation-guided network (SCGNet) to overcome the mentioned limitations. It consists of a semantic branch and a change branch, between which continuous information interaction is established to enable collaborative information utilization and joint optimization. SCGNet integrates three key modules: a semantic-guided fusion (SGF) module exploits semantic relevance cues from the semantic branch to guide adaptive bitemporal feature fusion in the change branch; a semantic consistency enhancement (SCE) module introduces invariant information extracted from the change branch to stabilize bitemporal semantic representations and enhance overall semantic consistency; and a multiscale change activation (MSCA) module activates and refines change features under semantic constraints across multiple scales, further strengthening the collaboration between semantic information and change responses. By coupling semantic reasoning and change perception through a continuous information interaction mechanism, SCGNet enables more reliable semantic change modeling. Positive experimental results counted on two benchmark datasets show that SCGNet outperforms many existing methods, confirming its effectiveness. Our source code is available at https://github.com/TangXu-Group/Semantic-Change-Detection/tree/main/SCGNet**

***Index Terms*—Remote sensing (RS), semantic change detection (SCD), semantic consistency.**

I. Introduction

BINARY change detection (BCD) is an important research topic in the remote sensing (RS) community, aiming to distinguish the changed areas from RS images covered the exact location but are captured at different times [1]. BCD plays an essential role in broad applications, such as postdisaster reconstruction [2], land use and land cover controlling [3], urban planning [4], and so on. However, in practical applications, researchers focus more on what exactly is changing rather than the mere occurrence of change. Therefore, semantic change detection (SCD) [5], [6], [7] appears. Compared to BCD, SCD not only needs to determine whether a change has occurred but also needs to complete semantic segmentation (SS) at the same time. Due to its granularity and applicability, SCD has received increasing attention.

Recently, with the development of deep learning, especially the convolutional neural network (CNN), many advanced methods have been proposed for diverse RS tasks [8], [9], [10], [11], [12], [13], [14], as well as SCD tasks [15], [16], [17]. These SCD methods can be roughly divided into three categories: direct classification [18], [19], postclassification comparison [20], [21], and multitask learning [22], [23]. Daudt et al. [24] compared and analyzed the above three methods. Experimental results show that multitask learning strategies perform best. This is because the former two are not good at modeling temporal semantic dependencies and cannot fully understand the semantic transformation process before and after the change. The latter can fully explore the intrinsic connections between semantic and change clues and increase the model's ability to distinguish semantic category conversions. Thus, the multitask learning paradigm is gradually becoming the mainstream research direction in SCD tasks [25], [26], [27].

The above SCD solutions have succeeded in their applications, however, there is still room for improvement. First, mainstream multitask learning frameworks simultaneously model SS and BCD tasks. Nonetheless, differences in optimization objectives between these two tasks lead to highly coupled feature learning processes, which would easily cause intertask interference. Existing methods often fail to adequately capture the intrinsic relationship between the two tasks. Therefore, it is necessary to construct a network architecture that can explicitly model the collaborative relationship between SS and BCD tasks. Second, RS image features often suffer from significant intraclass variations and minor interclass differences. This becomes even more pronounced in the semantic space of bitemporal images. However, most existing studies attempt to alleviate it primarily from the perspective of prediction losses or supervision constraints [22], [28], while paying insufficient attention to modeling the consistency

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Xu Tang, Weihao Jiang, Jingjing Ma, Xiangrong Zhang, and Licheng Jiao are with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, School of Artificial Intelligence, Xidian University, Xi'an 710071, China (e-mail: tangxu128@gmail.com; jjma@ieee.org).

Fang Liu is with the Key Laboratory of Intelligent Perception and Systems for High-Dimensional Information, Ministry of Education, School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China.

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TABLE I
COMPARISON BETWEEN SCGNET AND EXISTING SCD FRAMEWORKS FROM THE PERSPECTIVE OF INTERACTION MECHANISMS

Method Category	Interaction Direction	Interaction Level	Interaction Pattern	Correlation Modeling
Semantic-guided methods	One-way	Feature-level	Limited	Partial
Consistency-based methods	None or weak	Loss-level	Indirect	Implicit
SCGNet (Ours)	Bidirectional	Feature-level	Continuous	Explicit

of bitemporal semantic representations themselves. Hence, developing an efficient modeling mechanism that can enhance bitemporal semantic consistency is of great significance. Here, semantic consistency means maintaining the coherence and alignment of semantic representations for the same category across different temporal images.

Based on the above discussion, this article proposes a new SCD method called the semantic correlation-guided network (SCGNet). It adopts a dual-branch architecture consisting of a semantic branch and a change branch, which are jointly optimized. To this end, three task-oriented modules are designed and integrated into the network. The semantic-guided fusion (SGF) module is embedded in the change branch. It exploits semantic relevance cues from the semantic branch to emphasize spatial locations with significant semantic discrepancies. This way, the dual-temporal features can be adaptively and fully fused. The semantic consistency enhancement (SCE) module is incorporated into the semantic branch, where invariant information derived from the change branch is used as a filtering signal. This design alleviates the adverse effects of intraclass variation and interclass ambiguity, leading to more robust semantic representations. Furthermore, a multiscale change activation (MSCA) module is introduced to reinforce the collaboration between the two branches from a multiscale perspective. By jointly activating change features under semantic constraints, MSCA promotes semantic-aware change modeling and improves the completeness of change representations. Through the coordinated operation of these modules, SCGNet achieves reliable semantic–change interaction and generates higher-quality semantic change maps.

To further clarify the differences between SCGNet and existing SCD methods, we summarize their interaction mechanisms in Table I. As shown in the table, existing methods typically associate SS and BCD via semantic guidance or consistency losses, yet are often limited to one-way guidance, loss-level constraints, or sporadic interactions, leading to implicit or insufficient modeling of cross-task relationships. In contrast, SCGNet establishes an explicit semantic–change collaborative modeling mechanism via continuous bidirectional feature-level interaction, thereby capturing the intrinsic dependence between the two tasks. In addition, from an optimization perspective, SCGNet reduces task interference by transforming the conventional loosely coupled optimization of SS and BCD into a conditionally dependent process through continuous bidirectional feature interaction. Specifically, each task is optimized with awareness of the other, enabling mutual perception of optimization objectives. As a result, gradients are implicitly modulated by cross-branch information, forming coconditioned gradients rather than conflicting ones. By jointly aligning optimization objectives and feature representations, SCGNet achieves more stable and coherent optimization, effectively reducing task interference.

The following summarizes the main contributions of our work.

1) Considering the characteristics of SCD, SCGNet is designed as a two-branch structure consisting of a semantic branch and a change branch that are jointly optimized. Through continuous information exchange, semantic cues guide reliable change reasoning, while change-aware information stabilizes semantic representations, enabling the two branches to complement each other and produce higher-quality results.
2) To enable more efficient and adaptive bitemporal feature fusion, SGF is designed for the change branch. It utilizes the bitemporal semantic relevance from the semantic branch to highlight regions with significant semantic differences between bitemporal RS images, thereby enhancing the perception of key change areas.
3) To address the challenge in RS of significant intraclass variability and subtle interclass distinctions, SCE is introduced into the semantic branch. By leveraging invariant information derived from the change branch as a filter, SCE strengthens the representation of unchanged regions, thus improving overall semantic consistency.
4) To further enhance bidirectional collaboration between the semantic branch and the change branch, MSCA is proposed. MSCA implicitly constrains the interaction between semantic information and change features across multiple scales, enabling semantic-aware change modeling and encouraging mutual reinforcement between the two branches.

The rest of this article is organized as follows. Section II describes the published BCD and SCD methods. Section III explains the proposed SCGNet in detail. The experiments and related discussion are exhibited in Section IV. Section V draws a short conclusion for this article.

II. RELATED WORK

A. Binary Change Detection

In recent years, CNN-based methods have dominated the BCD community. In the beginning, since the BCD task can be regarded as a two-class classification problem, U-Net [29], which is popular in SS, attracted the researchers' eyes. Based on U-Net, Daudt et al. [30] proposed three networks, named FC-EF, FC-Siam-conc, and FC-Siam-diff. FC-EF leverages a single encoder to learn the knowledge from bitemporal RS images simultaneously, while FC-Siam-conc and FC-Siam-diff adopt a Siamese encoder to extract valuable

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information from different RS images. Through comprehensive experimental analysis, the beneficial effects of UNet and the Siamese encoder framework on BCD have been validated. Subsequently, Siamese networks have been proposed under the paradigm of U-Net [31], [32], [33], significantly improving BCD results. To further improve the performance of CNN-based methods, various advanced technologies have been added to them. For instance, a dual-branch multilevel intertemporal network (DMINet) was proposed in [34]. It unifies the self-attention and cross-attention mechanisms into a single module to learn features that are robust against external disturbances from RS images. Satisfactory change maps can be achieved along with a simple CNN backbone. For another example, He et al. [35] introduced dilated convolutions of different scales to obtain multiscale information to help cover objects of arbitrary shapes and sizes, thereby capturing various changes from macro to details and improving BCD results.

With the appearance and the development of visual Transformers [36], [37], [38], scholars use them to complete BCD tasks due to their powerful global information extraction ability. Zhang et al. [39] proposed SwinSUNet based on Swin Transformer [37]. It mainly includes an encoder, a fusion module, and a decoder. All of them are composed of the Swin Transformer block, which can extract global information from bitemporal RS images. The well BCD results confirm the usefulness of the global clues and Transformers. However, pure-Transformer architecture leads to high computational costs and memory requirements. To mitigate these problems, many profitable schemes are proposed. For instance, Ma et al. [40] proposed a method called the edge-assisted adaptive transformer detector. It introduces a self-adaptive vision transformer, which effectively reduces the computational burden by compressing the spatial resolution of feature maps and reconstructing them. At the same time, an edge-aware decoder is designed to refine the features further. Finally, promising results are obtained. Considering CNNs and Transformers focus on different aspects during the feature learning, the researchers combine them for addressing BCD tasks [41], [42], [43]. For example, Tang et al. [44] developed a method named WNet by adopting CNN and Transformer in a parallel structure. It introduces the concept of deformability in two backbones to understand irregular changes. Also, WNet proposes an improved channel attention mechanism for the fusion of two types of information. The discriminative and complete features bring satisfactory detection results. Another method, named AMTNet, was developed in [45]. It first uses a CNN as the backbone with a peculiar information exchange scheme to learn the local features and bridge the domain gaps between different temporal RS images. Then, a Transformer is adopted to encode and decode the local features to obtain robust features for BCD.

B. Semantic Change Detection

Over the past few years, many SCD methods have been developed based on deep learning techniques, achieving remarkable results [19], [21], [26]. They can be roughly divided into three groups, including direct classification methods, postclassification comparison methods, and multitask learning methods.

The direct classification methods take bitemporal RS images as input and directly output each pixel's "from–to" semantic change category. For example, Mou et al. [18] combined CNN and recurrent neural network (RNN) to generate high-quality semantic change maps. The former generates spatial feature representation, while the latter effectively analyzes temporal dependencies. Although feasible, treating the change type as a class label would lead to serious class imbalance problems. The method in [19] adopts a global stratified sampling mechanism to alleviate these issues. The positive semantic change maps can be produced along with the change mask strategy, which is good at weakening the background influence of the unchanged area.

The postclassification comparison methods first perform SS on the bitemporal RS images, respectively, and then achieve the changed areas and their semantic labels through postprocessing. For example, the method in [20] first learns the difference features from two SS branches. Then, the pleasurable SCD results can be obtained with the assistance of multiscale analysis, channel attention mechanism, and deep supervision technologies. While the above method yields good results, there is an issue we cannot ignore, i.e., the incorrect predictions of the two branches will accumulate and affect the detection results. To handle this trouble, Xia et al. [21] developed a new method to improve land cover classification accuracy by modeling temporal correlations. Then, a soft fusion strategy is adopted to determine the change types adaptively. By doing so, the error accumulation can be whittled down to enhance the SCD results.

The multitask learning methods treat SS and BCD as two complementary tasks for joint learning, setting up semantic prediction and change detection branches, respectively, based on a shared feature extraction network. Ding et al. [22] designed an SCD model, named Bi-SRNet. It consists of two semantic branches and one change branch. To better analyze temporal semantic features, a twin semantic reasoning block, and a cross-cycle semantic reasoning block are introduced. In addition, a new loss function is proposed to enhance semantic consistency. Satisfactory experimental results verify the ability of Bi-SRNet. In [23], an adaptive change enhancement method was proposed to help locate the change areas and utilize the change information to guide further semantic branching. The interaction between branches helps the network achieve satisfactory results. To model complex temporal or spatial relationships over long distances, Transformers and their variants have become increasingly popular. Zheng et al. [25] proposed a temporal-symmetric transformer and embedded it into an encoder–decoder structure. It aims to enhance the spatiotemporal interaction between bitemporal features. The achieved highly expressive features allow satisfactory results even with a simple decoder. Similarly, SCanNet [26] inputs the semantic features and change features into the Transformer to help model the spatiotemporal dependence. Also, it generates pseudolabels based on the bitemporal semantic distance to guide the development of semantic information. Extensive experiments verified its superior performance. Unlike the

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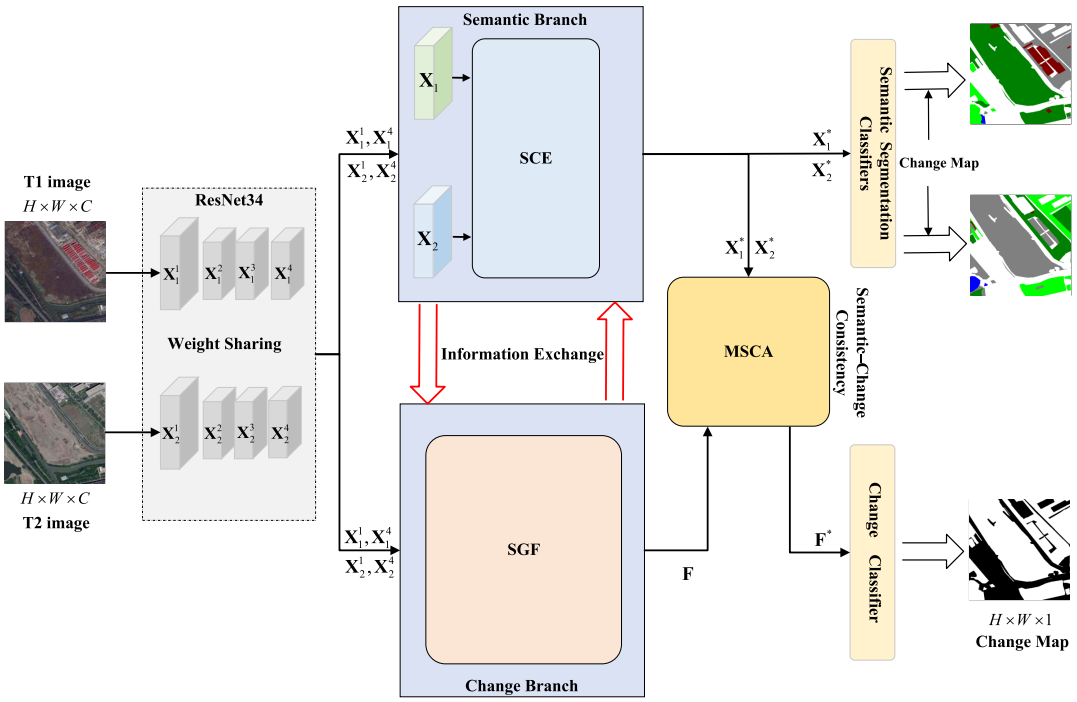


Fig. 1. Framework of the proposed SCGNet. It mainly contains a feature extractor, a semantic branch, a change branch, and an MSCA module. The feature extractor is used to learn the multiscale information from bitemporal RS images. The semantic branch, which includes a SCE module, focuses on learning stable semantic representations by exploiting invariant information. The change branch incorporates an SGF module to explore change cues under semantic guidance effectively. In addition, the MSCA module is introduced as a semantic–change consistency component to facilitate collaborative interaction between the two branches. By combining SS and change maps, final semantic change maps can be generated.

methods mentioned above that use Transformer as a model component, other works use Transformer as a backbone for global feature extraction. For example, Long et al. [46] designed a multilevel perceptual aggregation network. Based on the Transformer, temporal correlation analysis, and difference analysis are performed on the extracted discriminative features to create reliable semantic change maps. In [28], a standard Transformer is used to extract features. Then, the bitemporal features are split along the channel dimension and progressively concatenated. Subsequently, 3-D convolutions are employed to explore the interactions between temporal features further. This operation effectively enhances the model's representation ability in change regions, leading to suitable performance. Recently, Mamba [47] has been introduced as an efficient alternative to Transformers. Chen et al. [48] proposed MambaSCD, which employs Mamba to perform SS and BCD within a multitask framework jointly. It enables effective spatiotemporal modeling with linear computational complexity and achieves a favorable tradeoff between computational cost and performance. In general, multitask learning networks can increase the collaborative constraints between tasks and improve the accuracy of change category recognition. Thus, this type of method has become the current mainstream SCD method design idea [49], [50], [51].

III. Methodology

A. Overall Architecture

The SCGNet framework is shown in Fig. 1. It is composed of a weight-shared feature extractor based on ResNet34 [52], a semantic branch, a change branch, and an MSCA module. The semantic branch is equipped with an SCE module, while the change branch incorporates an SGF module. The MSCA module is introduced as a branch-consistency component. Given a pair of bitemporal RS images $\{\mathbf{I}_1, \mathbf{I}_2\} \in \mathbb{R}^{H\times W\times C}$, the feature extractor first learns the multiscale information from them. After obtaining four pairs of feature maps corresponding to different scales, we select the features of the shallowest layer (rich in texture information) and the deepest layer (rich in semantic information) for subsequent SCD, denoted as $\{\mathbf{X}_1^1, \mathbf{X}_2^1\} \in \mathbb{R}^{H/4\times W/4\times C_1}$ and $\{\mathbf{X}_1^4, \mathbf{X}_2^4\} \in \mathbb{R}^{H/8\times W/8\times C_2}$, where $C_1 = 64$ and $C_2 = 128$. It is worth noting that a slight modification is made to ResNet34, that is, the downsamplings are removed after the second residual stage. Then, the multiscale features are fed into two branches simultaneously. During this process, the two branches learn jointly and influence each other. As a result, both SGF and SCE naturally involve operations that incorporate information from the other branch. In detail, guided by bitemporal semantic relevance information from the semantic branch, SGF thoroughly mines the change clues hidden in the multiscale features and generates the change feature $\mathbf{F} \in \mathbb{R}^{H/4\times W/4\times C_2}$. SCE aims to discover the semantics within RS images on account of the invariant clues from the change branch. To this end, both semantic and change features are used to achieve the enhanced semantic features $\{\mathbf{X}_1^*, \mathbf{X}_2^*\} \in \mathbb{R}^{H/4\times W/4\times C_2}$. Next, to further facilitate collaboration between the two branches, MSCA is applied to both semantic and change features. By implicitly modeling the consistency between semantic differences and change

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responses in the multiscale feature space, MSCA refines the change representation and produces the enhanced change feature $\mathbf{F}^* \in \mathbb{R}^{H/4\times W/4\times C_2}$. Finally, with the help of SS and change classifiers, the SS maps and BCD change map can be achieved using $\mathbf{X}_1^*$, $\mathbf{X}_2^*$, and $\mathbf{F}^*$. After the simple multiplication, the final semantic change map can be generated.

B. SGF Module

As mentioned above, SGF aims to explore the change clues from bitemporal RS images. Meanwhile, semantic relevance information from the semantic branch is also used to guide adaptive fusion of bitemporal features, enabling the model to characterize complex and heterogeneous change patterns better. To this end, a semantic relevance attention (SRA) mechanism is first developed, which is committed to highlighting the regions with significant semantic differences in bitemporal features. Then, SGF is constructed based on SRA.

SRA contains three key components: a weight-shared convolution layer, a cosine similarity calculation block, and a normalized mask generation block. Given a pair of feature maps $\{\mathbf{X}_1^i, \mathbf{X}_2^i\}$, SRA first applies a weight-shared 1×1 convolution to them for projecting them into a unified feature space. This step helps SRA achieve semantic alignment and reduces the number of channels. Next, the cosine similarity calculation block is used to calculate a preliminary similarity map between the two features. The value range of the current similarity map is between $[-1, 1]$, which is not conducive to the network focusing on areas with large semantic differences. Thus, the normalized mask generation block is adopted to map the similarity map to a single-channel semantic change prompt map $\mathbf{SMask}^i$ with a value range of $[0, 1]$, which indicates the degree of attention that should be given to each area. The above SRA process can be expressed as

$$\begin{aligned} \mathbf{X}_1^i &= f_{conv1\times1}\left(\mathbf{X}_1^i\right), \mathbf{X}_2^i = f_{conv1\times1}\left(\mathbf{X}_2^i\right) \\ \cos_{sim}\left(\mathbf{X}_1^i, \mathbf{X}_2^i\right) &= \frac{\mathbf{X}_1^i \cdot \mathbf{X}_2^i}{\|\mathbf{X}_1^i\|\|\mathbf{X}_2^i\|} \\ \mathbf{SMask}^i &= \frac{1-\cos_{sim}\left(\mathbf{X}_1^i, \mathbf{X}_2^i\right)}{2} \end{aligned} \tag{1}$$

where $\cos_{sim}(\cdot)$ and $f_{conv1\times1}(\cdot)$ represent the cosine similarity function and 1×1 convolution, respectively. SRA leverages semantic relevance information to suppress false responses and highlight real change regions, thereby improving the localization accuracy and representation capability of the change branch.

Aided by SRA, we construct SGF here, whose structure is shown in Fig. 2. SRA is first used to generate a $\mathbf{SMask}^i$ to indicate the potential change areas. At the same time, $\mathbf{X}_1^i$ and $\mathbf{X}_2^i$ are concatenated in the channel dimension to obtain the feature map $\mathbf{X}_{cat}^i$, which retains the rich information in the bitemporal RS images, but also contains considerable redundancy. To filter out the redundant clues, we multiply $\mathbf{SMask}^i$ by $\mathbf{X}_{cat}^i$, and pass the result through a 3×3 convolutional layer to generate the initial fusion feature $\mathbf{F}_{in}^i$. This operation suppresses the interference of irrelevant areas and helps SGF adaptively fuse the features of changing areas. Subsequently, we perform element-by-element absolute difference operation

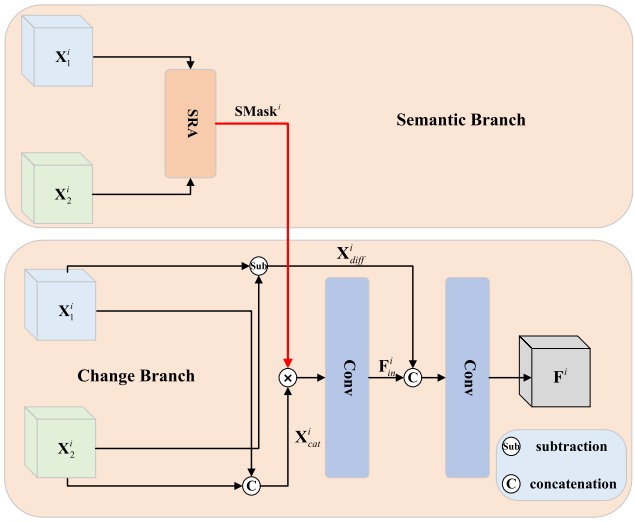


Fig. 2. Structure of the SGF module. It focuses on pixel locations with significant semantic differences under the guidance of semantic relevance information from the semantic branch, thereby enabling adaptive fusion of bitemporal features.

on $\mathbf{X}_1^i$ and $\mathbf{X}_2^i$ to obtain the difference feature map $\mathbf{X}_{diff}^i$. $\mathbf{X}_{diff}^i$ contains useful explicit change messages, but its materials are incomplete and unclear enough due to the lack of context and semantic support. In contrast, $\mathbf{F}_{in}^i$ provides stronger semantic completeness and regional consistency. Therefore, we concatenate $\mathbf{X}_{diff}^i$ and $\mathbf{F}_{in}^i$ in the channel dimension to make full use of their complementarity and further improve the expression of the changed regions. Finally, a 3×3 convolution layer is employed to get the fused feature $\mathbf{F}_i$. The above process can be expressed as

$$\begin{aligned} \mathbf{X}_{cat}^i &= \text{Concat}\left(\mathbf{X}_1^i, \mathbf{X}_2^i\right) \\ \mathbf{F}_{in}^i &= f_{conv3\times3}\left(\mathbf{X}_{cat}^i \cdot \mathbf{SMask}^i\right) \\ \mathbf{X}_{diff}^i &= |\mathbf{X}_1^i - \mathbf{X}_2^i| \\ \mathbf{F}^i &= f_{conv3\times3}\left(\text{Concat}\left(\mathbf{F}_{in}^i, \mathbf{X}_{diff}^i\right)\right) \end{aligned} \tag{2}$$

where $|\cdot|$ represents the absolute difference function, $f_{conv3\times3}(\cdot)$ denotes the 3×3 convolution, and $\text{Concat}(\cdot)$ means the channelwise concatenation. It is worth noting that the above description pertains to how SGF processes single-scale features. The final change feature $\mathbf{F}$ is obtained after multiscale fusion through the following equation:

$$\begin{aligned} \mathbf{F}^4 &= f_{conv1\times1}\left(\text{up}\left(\mathbf{F}^4\right)\right) \\ \mathbf{F} &= f_{conv3\times3}\left(\text{Concat}\left(\mathbf{F}^4, \mathbf{F}^1\right)\right). \end{aligned} \tag{3}$$

C. Semantic Consistent Enhancement Module

After integrating bitemporal features via SGF and obtaining a high-quality change feature $\mathbf{F}$ for BCD, we now focus on the SS subtasks for SCD. In SS, bitemporal RS images often exhibit significant differences in the semantic space, whittling down the network's discriminative ability. Therefore, enhancing the semantic consistency between bitemporal semantic branches has become an urgent problem. However, SCD tasks also focus on semantic changes in the change

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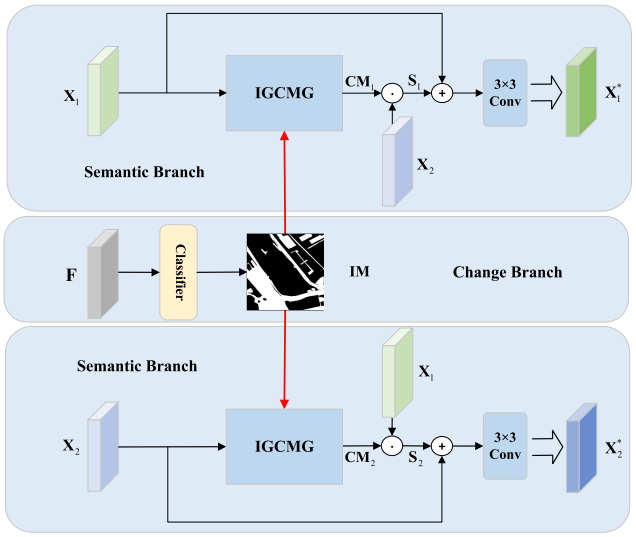


Fig. 3. Structure of the SCE module. It aims to utilize invariant information from the change branch for filtering guidance and enhance the representation of invariant regions in bitemporal RS images, thereby improving overall semantic consistency.

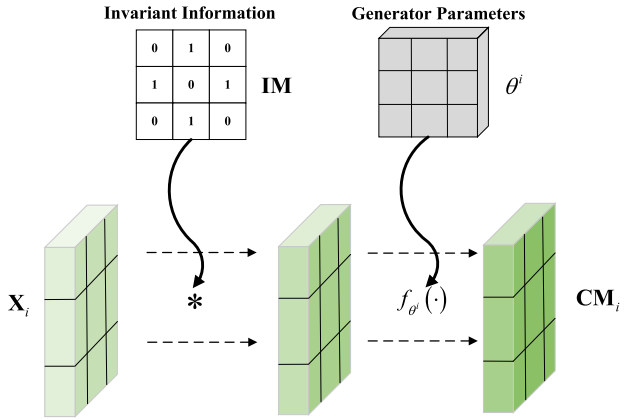


Fig. 4. Structure of the IGCMG mechanism. It generates the control matrix by jointly exploiting the single-temporal semantic feature, invariant information map, and generator parameters.

areas, and the semantics of these areas are inherently different, which is fundamentally at odds with improving semantic consistency. Fortunately, there are still many invariant areas in SCD, and their bitemporal semantics remain consistent. These invariant areas can, therefore, serve as trustworthy anchors for guiding semantic feature enhancement. Motivated by the above observation, we propose an SCE module, whose structure is illustrated in Fig. 3. SCE explicitly uses invariant information extracted from the change branch to guide semantic feature modulation, aiming to enhance semantic consistency in unchanged regions while suppressing unreliable semantic disturbances.

Specifically, when SCE receives the input data $\{\mathbf{X}_1, \mathbf{X}_2\} \in \mathbb{R}^{H/4\times W/4\times C_2}$ (cascaded by the upsampled $\mathbf{X}_i^4$ and $\mathbf{X}_i^1$, where i denotes the temporal phase) and $\mathbf{F}$, it first passes the change feature $\mathbf{F}$ through a simple classifier (consists of a 1×1 convolution layer and a sigmoid function) to generate a preliminary change prediction map, from which we further derive the invariant information map **IM**. Here, **IM** serves as a spatial prior that indicates unchanged regions and is further used to guide cross-temporal semantic interaction. Under the guidance of **IM**, SCE enables bitemporal semantic features to exchange reliable information across time, thereby promoting semantic consistency modeling in invariant areas. To realize invariant-guided semantic interaction flexibly and adaptively, we design a dedicated mechanism, invariant-guided control matrix generation (IGCMG), which works as the core component of SCE.

As illustrated in Fig. 4, IGCMG jointly exploits single-temporal semantic feature $\mathbf{X}_i$, the invariant information map **IM**, and the generator parameters θ^i to produce a control matrix $\mathbf{CM}_i$ for each temporal phase. The generated control matrix performs spatially adaptive dynamic filtering on the semantic features corresponding to different temporal phases, thereby enhancing semantic consistency in invariant regions while suppressing unreliable semantic disturbances. The generation process is formulated as

$$\mathbf{CM}_i = f_{\theta^i}(\mathbf{X}_i \cdot \mathbf{IM}), \quad i = 1, 2 \tag{4}$$

where $f_{\theta^i}(\cdot)$ denotes a 1×1 convolution followed by a sigmoid activation.

Using $\{\mathbf{CM}_1, \mathbf{CM}_2\}$ obtained by IGCMG, SCE performs invariant-guided cross-temporal semantic filtering to derive $\{\mathbf{S}_1, \mathbf{S}_2\}$, which encode semantic consistency information of invariant regions. After obtaining $\mathbf{S}_i$, it would be added to the corresponding original feature $\mathbf{X}_i$ in a residual manner to strengthen the expressions of the unchanged areas further. Finally, the two enhanced features are deeply modulated through 3×3 convolutions to obtain the robust semantic features $\mathbf{X}_1^*$ and $\mathbf{X}_2^*$. The overall SCE process can be formulated as

$$\begin{aligned} \mathbf{IM} &= f_c(\mathbf{F}) \\ \mathbf{CM}_i &= \mathrm{IGCMG}(\mathbf{X}_i, \mathbf{IM}), \quad i = 1, 2 \\ \mathbf{S}_1 &= \mathbf{CM}_1 \times \mathbf{X}_2, \quad \mathbf{S}_2 = \mathbf{CM}_2 \times \mathbf{X}_1 \\ \mathbf{X}_i^* &= f_{conv3\times3}(\mathbf{X}_i + \mathbf{S}_i), \quad i = 1, 2 \end{aligned} \tag{5}$$

where $f_c(\cdot)$ denotes the classifier used to derive the invariant information map, IGCMG$(\cdot)$ indicates the function of the IGCMG mechanism, and $f_{conv3\times3}(\cdot)$ denotes the 3×3 convolution.

D. MSCA Module

When the enhanced semantic features $\mathbf{X}_1^*$ and $\mathbf{X}_2^*$, and the change feature $\mathbf{F}$ are obtained, we further focus on how to promote bidirectional collaboration between the semantic and change branches. Therefore, the MSCA module is introduced, which models the semantic–change relationships from a multiscale spatial perspective. It enhances the change feature by injecting semantically guided spatial responses into change representations, thereby promoting coherent semantic–change predictions collaboratively. The overall architecture of MSCA is illustrated in Fig. 5, which consists of an implicit semantic–change consistency activation part and a multiscale feature aggregation part.

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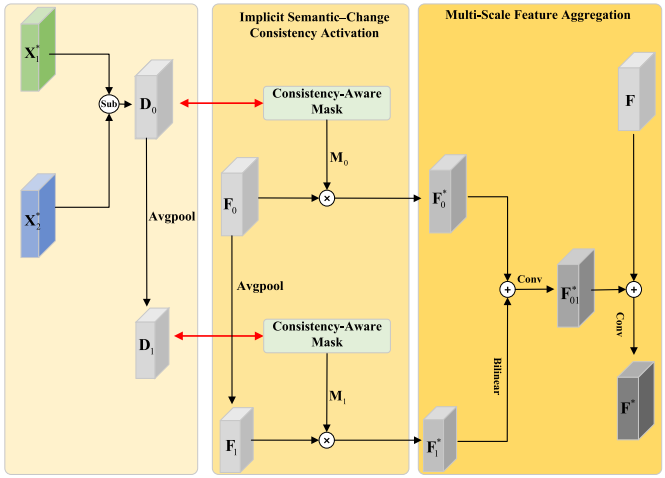


Fig. 5. Flowchart of the MSCA module. Semantic discrepancies derived from bitemporal semantic representations are mapped to consistency-aware masks, which implicitly modulate change features in the spatial domain. By aggregating multiscale features, MSCA enhances change representations while promoting spatial coherence between semantic transitions and change responses. For clarity, only two scales are illustrated.

Specifically, let $\mathbf{D}$ denote the semantic difference feature map obtained by $\mathbf{X}_1^*$ and $\mathbf{X}_2^*$. To transfer semantic cues into the change branch and establish semantic–change consistency, $\mathbf{D}$ is first projected into a single-channel consistency-aware mask $\mathbf{M}$ through a 1×1 convolution followed by a sigmoid activation. This operation converts semantic discrepancy information into a spatially adaptive modulation signal. Subsequently, $\mathbf{M}$ is applied to $\mathbf{F}$ via elementwise multiplication. This operation strengthens regions where semantic discrepancy is highly consistent with change patterns, while effectively suppressing responses that are semantically mismatched with the change representation. On this basis, the modulated change features further influence the stability and reliability of semantic predictions during collaborative optimization, thereby forming an implicit cross-branch bidirectional feedback mechanism. The above process can be regarded as an implicit semantic–change consistency activation mechanism, which constitutes the foundation of MSCA.

To further account for spatial diversity and scale-dependent semantic–change correspondence, the above process is extended to a multiscale setting. Specifically, average pooling operations are employed to generate multiscale representations of $\mathbf{F}$ and $\mathbf{D}$, denoted as $\{\mathbf{F}_i, \mathbf{D}_i | i = 0, 1, 2\}$, where i indicates the scale level. The spatial sizes of these feature maps are $h^i \times w^i \times C_2$, with $h^i = H/2^{i+2}$ and $w^i = W/2^{i+2}$. At each scale, semantic–change consistency is modeled independently through spatial modulation. Taking the ith scale as an example, the implicit semantic–change consistency activation is formulated as

$$\mathbf{M}_i = \sigma\left(f_{conv1\times1}\left(\mathbf{D}_i\right)\right), \quad \mathbf{F}_i^* = \mathbf{F}_i \times \mathbf{M}_i \tag{6}$$

where $\sigma(\cdot)$ represents the sigmoid function. In this manner, each scale learns a scale-specific consistency-aware modulation pattern, allowing the network to capture semantic–change correspondence across different spatial resolutions. After obtaining the enhanced multiscale change features $\{\mathbf{F}_0^*, \mathbf{F}_1^*, \mathbf{F}_2^*\}$, they are progressively fused from deeper to shallower levels to aggregate semantic–change cues across scales, which can be expressed as

$$\begin{aligned}
\mathbf{F}_{12}^* &= f_{conv1\times1}\left(\text{Bilinear}\left(\mathbf{F}_2^*\right) + \mathbf{F}_1^*\right) \\
\mathbf{F}_{012}^* &= f_{conv1\times1}\left(\text{Bilinear}\left(\mathbf{F}_{12}^*\right) + \mathbf{F}_0^*\right) \\
\mathbf{F}^* &= f_{conv3\times3}\left(\mathbf{F}_{012}^* + \mathbf{F}\right)
\end{aligned} \tag{7}$$

where Bilinear$(\cdot)$ denotes the bilinear interpolation function. By gradually refining and integrating the semantic–change clues at different levels, MSCA can effectively enhance the model's perception of the change region and ultimately achieve an accurate characterization of the region's characteristics.

E. Classifier and Loss Function

We add three simple classifiers on top of SCGNet, including two identical SS classifiers and a change classifier. The SS classifier consists of a 1×1 convolutional layer, and the change classifier consists of two 1×1 convolutional layers. After passing $\mathbf{X}_1^*$, $\mathbf{X}_2^*$, and $\mathbf{F}^*$ through different classifiers, we get the SS map $\mathbf{P}_s$ ($\mathbf{P}_1$ and $\mathbf{P}_2$), and the BCD change map $\mathbf{P}_c$. The final semantic change map can be obtained by using the $\mathbf{P}_c$ as a mask to multiply the $\mathbf{P}_s$.

To train SCGNet, we use three loss functions, namely a semantic loss (i.e., cross-entropy loss) $\mathcal{L}_{sem}$ [53], a change loss (i.e., binary cross-entropy loss) $\mathcal{L}_{change}$ [54], and a semantic consistency loss $\mathcal{L}_{sc}$ [22]. The semantic loss is responsible for the semantic branch, the change loss is for the change branch, and the semantic consistency loss improves the semantic expression ability of SCGNet by encouraging similar semantic predictions in unchanged regions while penalizing overly consistent predictions in changed areas. Suppose the SS ground truths (GTs) are $\mathbf{G}_s$ ($\mathbf{G}_1$ and $\mathbf{G}_2$), and the BCD GT is $\mathbf{G}_c$. The definitions of $\mathcal{L}_{sem}$, $\mathcal{L}_{change}$, and $\mathcal{L}_{sc}$ are

$$\begin{aligned}
\mathcal{L}_{sem} &= -\frac{1}{N}\sum_{i=1}^{N} y_i \log\left(p_i\right) \\
\mathcal{L}_{change} &= -y_c \log\left(p_c\right) - \left(1 - y_c\right)\log\left(1 - p_c\right) \\
\mathcal{L}_{sc} &= \begin{cases} 1 - \cos\left(x_1, x_2\right), & y_c = 1 \\ \cos\left(x_1, x_2\right), & y_c = 0 \end{cases}
\end{aligned} \tag{8}$$

where N represents the number of categories, y_i and p_i indicate the corresponding labels and predicted probabilities in $\mathbf{G}_s$ and $\mathbf{P}_s$, respectively, y_c and p_c denote the corresponding labels and predicted probabilities in $\mathbf{G}_c$ and $\mathbf{P}_c$, respectively, $\cos(\cdot)$ implies the cosine distance, and x_1 and x_2 are the feature vectors of pixels on $\mathbf{P}_1$ and $\mathbf{P}_2$, respectively. Finally, the overall loss function formula of our SCGNet can be defined as

$$\mathcal{L} = \left(\mathcal{L}_{sem1} + \mathcal{L}_{sem2}\right)/2 + \mathcal{L}_{change} + \mathcal{L}_{sc}. \tag{9}$$

IV. Experiment

A. Datasets Introduction

We select two popular datasets in SCD to verify our proposed SCGNet. One is SECOND [16], and the other is Landsat–SCD [55]. SECOND includes 4662 RS image

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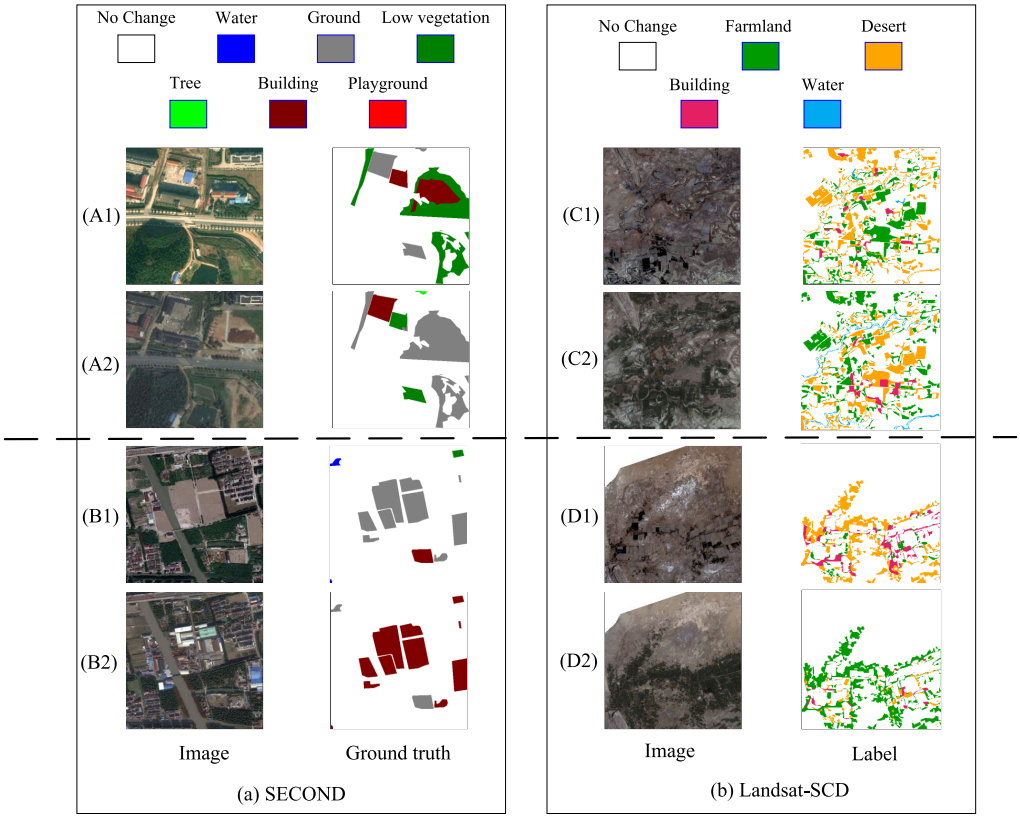


Fig. 6. Examples of bitemporal RS images and GT from SECOND and Landsat–SCD. (a) Two RS image pairs and their GTs from SECOND, which are recorded as (A1)/(A2) and (B1)/(B2). (b) Two RS image pairs and their GTs from Landsat–SCD, which are denoted by (C1)/(C2) and (D1)/(D2).

pairs obtained from various cities in China. Their spatial sizes are 512×512, and their spatial resolution varies from 0.5 to 3 m. Among them, only 2968 pairs of RS images are publicly available, and their GT labels consist of seven categories: unchanged, nonvegetated ground surface, tree, low vegetation, water, buildings, and playgrounds. According to [22], these labeled RS image pairs are randomly divided into training and test sets with a ratio of 4:1. In other words, 2375 pairs are used for training, and 593 pairs are used for testing. The examples of bitemporal RS images and their GTs of SECOND are shown in Fig 6(a). Landsat–SCD was constructed using the RS images of the desert edge in Xinjiang, China, obtained between 1990 and 2020. There are 2425 original RS image pairs in Landsat–SCD, and their spatial sizes and spatial resolution are 416×416 and 30 m, respectively. To enrich the dataset, the researchers augment the RS images using normal operations, such as flipping, masking, and resizing. Finally, the number of image pairs reaches 8468. The GTs of this dataset include five categories: unchanged, farmland, desert, building, and water. As suggested in [26], we select the original 2425 pairs and further divide them into training and test sets with a numerical ratio of 4:1 (i.e., 1940 and 485 pairs, respectively). Fig. 6(b) presents examples of RS image pairs and their GTs of Landsat–SCD.

B. Experimental Settings

All experiments are conducted on a server equipped with one NVIDIA GeForce RTX 4090 graphics card, featuring 24 GB of video memory, utilizing a Python environment and a PyTorch compilation platform. In the training process, random flipping [56] is used to augment the RS images, ensuring the stability of the network. The parameters of different parts of our SCGNet are initialized randomly except for the ResNet34 backbone, whose parameters are initialized using pretraining weights [52]. The SGD optimizer [57] is used to optimize SCGNet. The learning rate is initialized as 0.1 and gradually decreases with the training iterations. For SECOND, the batch size is set to 8, and the training is conducted for 50 epochs. For Landsat–SCD, the batch size is also set to 8 with 100 training epochs.

Four assessment criteria are selected to measure SCGNet's performance numerically, including overall accuracy (OA), mean intersection over union (mIoU), separated kappa (SeK) coefficient, and SCD-targeted $F1$ score (F_{scd}). OA represents the proportion of correctly classified pixels in the whole. Let us use q_{ij} to represent the number of pixels classified as class i whose GT index is j. Then, OA can be calculated as follows:

$$\mathrm{OA} = \sum_{i=0}^{N} q_{ii} \Big/ \sum_{i=0}^{N} \sum_{j=0}^{N} q_{ij} \tag{10}$$

where N represents the number of categories. mIoU measures the average intersection over the union ratio of prediction and GT in all categories, and its definition is

$$\mathrm{mIoU} = (\mathrm{IoU}_{\mathrm{nc}} + \mathrm{IoU}_{\mathrm{c}})/2$$

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TABLE II
COMPARISON RESULTS COUNTED ON TWO DATASETS. THE BEST VALUES ARE IN BOLD, AND THE SECOND-BEST VALUES ARE UNDERLINED

Methods	SECOND				Landsat-SCD			
	OA(%)	mIoU(%)	Sek(%)	F_{scd}(%)	OA(%)	mIoU(%)	Sek(%)	F_{scd}(%)
HRSCD-str.3	83.51	65.29	10.49	49.78	90.00	76.06	26.44	69.36
HRSCD-str.4	85.77	71.00	17.45	56.92	90.75	77.94	29.93	71.59
SSCD-l	87.41	72.66	22.39	61.80	94.15	83.46	45.40	82.75
Bi-SRNet	87.71	73.17	22.96	62.22	94.46	84.28	47.71	83.83
MambaSCD	87.43	73.20	23.21	62.62	95.68	87.84	55.97	88.23
EGMS-Net	86.93	73.04	23.35	62.71	96.56	90.76	66.38	91.09
CdSC	87.59	72.98	23.01	62.56	94.78	85.32	49.62	84.34
HGINet	87.05	72.42	22.08	61.88	94.81	85.55	50.26	84.65
SCanNet	**87.78**	73.27	23.11	62.81	96.32	89.34	61.39	89.54
SCGNet(ours)	87.74	**73.66**	**24.04**	**63.42**	**97.10**	**91.13**	**67.06**	**91.56**

$$\mathrm{IoU}_{nc} = q_{00} \Big/ \left(\sum_{i=0}^{N} q_{i0} + \sum_{j=0}^{N} q_{0j} - q_{00} \right)$$
$$\mathrm{IoU}_{c} = \sum_{i=1}^{N}\sum_{j=1}^{N} q_{ij} \Big/ \left(\sum_{i=0}^{N}\sum_{j=0}^{N} q_{ij} - q_{00} \right) \tag{11}$$

where $\mathrm{IoU_{nc}}$ and $\mathrm{IoU_c}$ represent the IoU of the unchanged area and the changed area, respectively. SeK evaluates the classification accuracy of semantic categories in the change areas. It can be formulated as

$$\mathrm{SeK} = e^{\mathrm{IoU}_c - 1} \cdot (\rho - \eta)/(1 - \eta)$$
$$\rho = \sum_{i=0}^{N} \hat{q}_{ii} \Big/ \sum_{i=0}^{N}\sum_{j=0}^{N} \hat{q}_{ij}$$
$$\eta = \sum_{i=0}^{N} \left(\sum_{j=0}^{N} \hat{q}_{ij} \cdot \sum_{j=0}^{N} \hat{q}_{ji} \right) \left(\sum_{i=0}^{N}\sum_{j=0}^{N} \hat{q}_{ij} \right)^{2} \tag{12}$$

where $\hat{q}_{ij}$ is equal to q_{ij}, except that $\hat{q}_{00}$ is set to 0, and ρ and η represent the proportion of correctly classified samples and the expected accuracy under random guessing that are computed only over changed regions, respectively. F_{scd} is obtained based on precision (P_{scd}) and recall (R_{scd}), reflecting the overall performance. The definition of F_{scd} is

$$F_{scd} = 2 \cdot P_{scd} \cdot R_{scd}/(P_{scd} + R_{scd})$$
$$P_{scd} = \sum_{i=1}^{N} q_{ii} \Big/ \sum_{i=1}^{N}\sum_{j=0}^{N} q_{ij}$$
$$R_{scd} = \sum_{i=1}^{N} q_{ii} \Big/ \sum_{i=0}^{N}\sum_{j=1}^{N} q_{ij}. \tag{13}$$

C. Performance of SCGNet

To verify the effectiveness of SCGNet, we select nine SCD methods for comparison, including HRSCD-str.3 [24], HRSCD-str.4 [24], SSCD-l [22], Bi-SRNet [22], MambaSCD [48], EGMS-Net [23], CdSC [28], HGINet [46], and SCanNet [26]. HRSCD-str.3, HRSCD-str.4, SSCD-l, Bi-SRNet, and EGMS-Net are CNN-based models, MambaSCD is based on Mamba, while the remaining models are developed based on the Transformer. All compared methods are completed using the codes released by the authors. Their free parameters are set according to the original article, and their experimental settings are consistent with the contents discussed in Section IV-B.

1) Numerical Results: The numerical results of different methods counted on the two datasets are shown in Table II, where the best values are in bold, and the second-best values are underlined.

Through observation, we find that all compared methods achieve acceptable performance on the SECOND dataset. However, HRSCD-str.3 and HRSCD-str.4 show relatively weaker results due to their simple structures, which are insufficient for SCD. The other compared methods yield similar performance. For instance, due to the incorporation of a pseudolabel learning mechanism, the semantic representation of SCanNet can be ensured, allowing it to achieve the highest OA values among all compared models. Similarly, the behavior of EGMS-Net is the strongest among all compared methods in terms of SeK since it focuses on highlighting the changed regions. Despite the good performance of the above methods, SCGNet still outperforms them in most cases. Taking F_{scd} as an example, SCGNet achieves performance improvements of 13.64% over HRSCD-str.3, 6.50% over HRSCD-str.4, 1.62% over SSCD-l, 1.20% over Bi-SRNet, 0.80% over MambaSCD, 0.71% over EGMS-Net, 0.86% over CdSC, 1.54% over HGINet, and 0.61% over SCanNet. For SeK, SCGNet yields enhancements of 13.55% over HRSCD-str.3, 6.59% over HRSCD-str.4, 1.65% over SSCD-l, 1.08% over Bi-SRNet, 0.83% over MambaSCD, 0.69% over EGMS-Net, 1.03% over CdSC, 1.96% over HGINet, and 0.93% over SCanNet.

The above positive results of SCGNet can be attributed to the following points. First, SGF enables the network to adaptively fuse bitemporal features and focus on the real change area. Second, SCE effectively helps SCGNet model the semantic consistency of bitemporal features. Even though SCGNet achieves pleasurable results on SECOND, we cannot ignore the following point, i.e., it does not perform best in OA. The reason behind this can be summarized as follows. Due to the special fusion mechanism of SGF and the reactivation of MSCA for the change areas, SCGNet pays more attention to the changed pixels. In other words, SCGNet tends to classify

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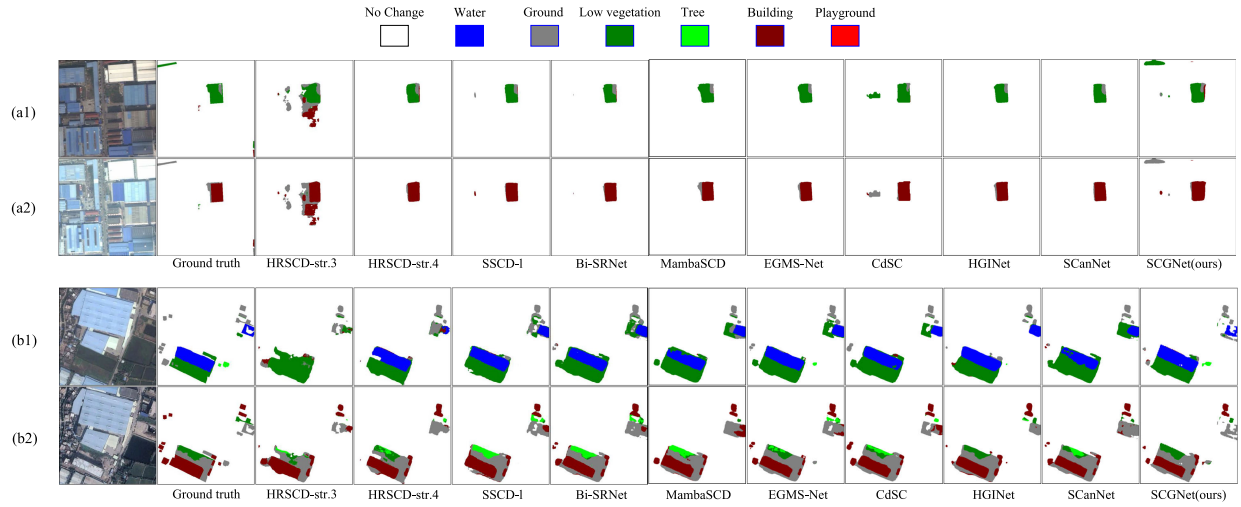


Fig. 7. Semantic change maps on SECOND, which are obtained by different methods. (a1)/(a2) and (b1)/(b2) are two RS image pairs that are selected from SECOND randomly. The GTs and semantic change maps achieved by various methods are displayed in order.

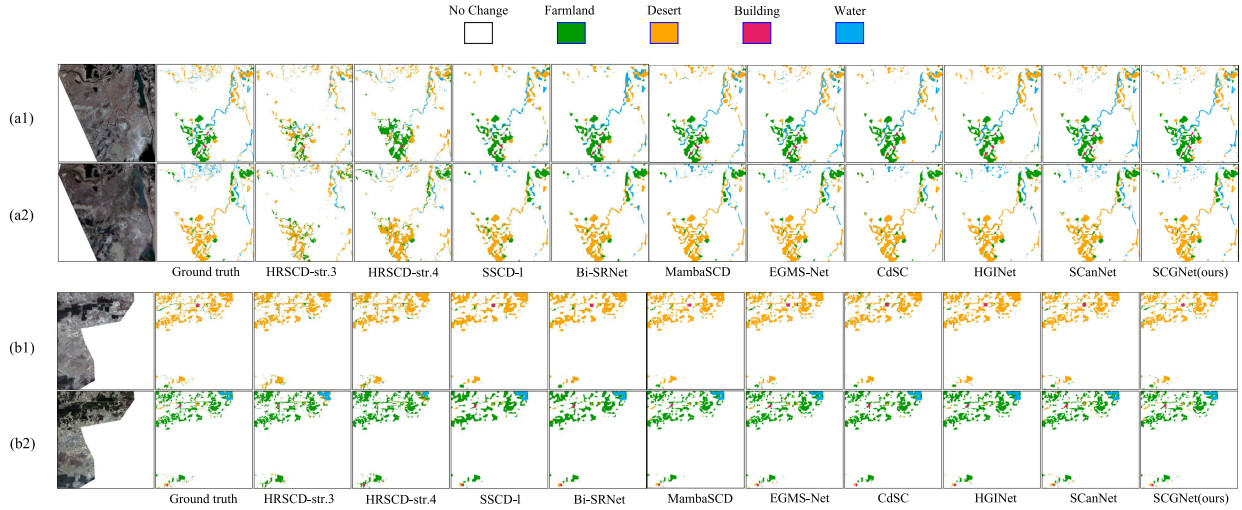


Fig. 8. Semantic change maps on Landsat–SCD, which are obtained by different methods. (a1)/(a2) and (b1)/(b2) are two RS image pairs that are selected from Landsat–SCD randomly. The GTs and semantic change maps achieved by various methods are displayed in order.

the pixels as changed. As a result, its OA value would be reduced more or less. Fortunately, the OA value of SCGNet still ranks in the top two, which underscores its usefulness in SCD.

Consistent findings are evident on the Landsat–SCD dataset. All methods achieve acceptable results, and SCGNet maintains the best performance on all evaluation metrics. An encouraging observation is that the performance gaps between SCGNet and other methods are distinct. For example, in OA, SCGNet surpasses the weakest compared method (i.e., HRSCD-str.3) by 7.10% and the strongest compared method (i.e., EGMS-Net) by 0.54%. Regarding mIoU, SCGNet outperforms HRSCD-str.3 by 15.07% and EGMS-Net by 0.37%. In addition to the factors mentioned above, the strong performance of our SCGNet also benefits from MSCA, which reinforces the representation of change regions by jointly modeling semantic–change cues at multiple scales, resulting in more precise and fine-grained change maps.

2) Visual Results: Apart from the numerical results, the quality of semantic change maps is also an important evaluation indicator. Therefore, we randomly select two RS image pairs from the two datasets and exhibit their semantic change maps, generated by SCGNet and other compared methods, in Figs. 7 and 8. By observing the visual results, it is not difficult to summarize the following advantages of SCGNet. First, SCGNet performs well in locating change areas, regardless of their size. In contrast, the performance of other methods is relatively inferior. For example, in the RS image pair (a1)/(a2) of Fig. 7, only SCGNet successfully identifies the small area in the top left where low vegetation has changed to ground, while other methods fail to detect the change in this area. This can be attributed to SGF, which adaptively fuses bitemporal features and effectively highlights the change regions. Second, SCGNet also shows obvious advantages in semantic classification tasks. For example, in the RS image pair (b1)/(b2) of Fig. 8, the low vegetation area on the bottom left is

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TABLE III
PARAMETER SIZE (PARAM) AND FLOATING-POINT OPERATIONS (FLOPS) OF DIFFERENT METHODS

Methods	Param (M)	FLOPs (G)
HRSCD-str.3	12.77	42.94
HRSCD-str.4	13.71	43.97
SSCD-l	23.31	189.76
Bi-SRNet	23.38	190.30
MambaSCD	21.51	73.42
EGMS-Net	23.92	216.09
CdSC	33.86	134.80
HGINet	27.70	50.64
SCanNet	27.90	264.95
SCGNet (Ours)	25.48	218.65

correctly classified by SCGNet. At the same time, most other models misclassify it as a similar tree category. This highlights the effectiveness of SCE, which enhances SCGNet's ability to model semantic consistency. Finally, the semantic change maps generated by SCGNet have clearer outlines, as shown in the two Landsat–SCD examples in Fig. 8. This advantage is credited to MSCA, which integrates semantic–change cues at multiple scales, strengthening the representation of change regions while further refining their spatial structures via multiscale pooling and feature reconstruction.

3) Model Complexity: To further study our SCGNet, we count its parameter size (Param) and floating-point operations (FLOPs) and display them in Table III. Here, the batch size and image size are equal to 1 and $512 \times 512 \times 3$, respectively. The compared methods' Param and FLOPs are also shown for reference. We can easily find that the lightest and heaviest models are HRSCD-str.3 and CdSC, while the models with the lowest and highest FLOPs are HRSCD-str.3 and SCanNet. For SCGNet, the parameter size is at a moderate level, while the FLOPs are relatively higher compared to some recent efficient models (e.g., Mamba-based approaches). This is mainly due to our backbone. As mentioned in Section III, a slight modification is applied to ResNet34 to meet the requirements of SCD tasks better: downsampling is removed after the second residual stage. Although this modification can preserve fine-grained spatial details and enhance high-level semantic representation, it inevitably increases computational cost. By comparison, the key modules proposed in this article (i.e., SGF, SCE, and MSCA) do not impose a significant computational burden. More details could be found in Section IV-D. Therefore, the performance improvement of SCGNet is mainly attributed to more effective cross-branch interaction rather than a significant increase in model complexity. In future work, we will further explore lightweight architectures under the proposed branch-collaborative modeling paradigm to achieve a better balance between performance and efficiency.

D. Ablation Studies

1) Contributions of Various Modules: In this section, we conduct ablation experiments on various modules of SCGNet to verify their contributions, including SGF, SCE, and MSCA. To this end, we first construct eight networks.

1) *Net-0:* Baseline.
2) *Net-1:* Baseline + SGF.
3) *Net-2:* Baseline + SCE.
4) *Net-3:* Baseline + MSCA.
5) *Net-4:* Baseline + SGF + SCE.
6) *Net-5:* Baseline + SCE + MSCA.
7) *Net-6:* Baseline + SGF + MSCA.
8) *Net-7:* Baseline + SGF + SCE + MSCA.

Net-0, denoted as the baseline, means that the semantic branch uses the deep features extracted by ResNet34 directly for semantic classification, and the change branch adopts channelwise concatenation as the bitemporal fusion method. Net-1, Net-2, and Net-3 add the SGF, SCE, and MSCA modules to Net-0, respectively, aiming to verify the effectiveness of these different modules. Net-4 combines SGF and SCE and embeds them into the baseline. Net-5 embeds SCE and MSCA into the baseline, and Net-6 integrates SGF and MSCA into the baseline. This way, the helpfulness of various combinations can be investigated. Net-7 accommodates the above three modules simultaneously, which is essentially the proposed SCGNet. For clarity, we adopt the SeK and F_{scd} scores as the evaluation metrics due to their comprehensive assessment ability. The Param(eters) and FLOPs are also counted for further study of different modules.

The results of eight networks counted on two datasets are exhibited in Table IV. It can be observed that Net-1 achieves performance improvements compared to Net-0. Taking SECOND as an example, the SeK and F_{scd} scores of Net-1 are 0.62% and 0.43% higher than those of Net-0. This illustrates that SGF helps bitemporal features merge and pushes the model to focus on real change pixels. Net-2 performs much better than Net-0. Taking Landsat–SCD as an example, the SeK and F_{scd} scores of Net-2 are 2.63% and 1.22% higher than those of Net-0. This is attributed to the assistance of SCE in modeling the semantic consistency. Net-3's behavior is also stronger than Net-0. Taking Landsat–SCD as an example, Net-3's SeK and F_{scd} scores are 1.86% and 0.67% higher than Net-0, respectively. This proves the effectiveness of implicitly modeling semantic–change consistency from a multiscale spatial perspective within MSCA for SCD.

For Net-4, Net-5, and Net-6, their performance is better than that of the above networks in most cases. This implies that the pairwise combination of different modules provides greater benefit to the baseline than any single module alone. In other words, combining modules with different functionalities enables the model to capture changing semantic information more accurately. The only exception is the performance of Net-6 on the SECOND dataset. When SGF and MSCA are used together, the scores on SeK and F_{scd} are 23.28% and 62.55%, respectively. This result is lower than the scores of 23.41% and 62.62% achieved by using SGF alone (i.e., Net-1). The reason for this is that in the semantically complex SECOND, MSCA uses features that have not been semantically enhanced to activate the change features. Since these semantic features contain more interference information, it is easy for the model to misjudge the changed areas, thus affecting the overall detection effect. Finally, Net-7 (i.e., SCGNet) has

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TABLE IV
ABLATION EXPERIMENTS IN SCGNET COUNTED ON TWO DATASETS. THE BEST VALUES ARE IN BOLD

Networks	SGF	SCE	MSCA	Param (M)	FLOPs (G)	SECOND		Landsat-SCD	
						Sek(%)	F_{scd}(%)	Sek(%)	F_{scd}(%)
Net-0	×	×	×	24.19	204.29	22.79	62.19	59.85	88.75
Net-1	✓	×	×	24.71	207.78	23.41	62.62	61.53	89.39
Net-2	×	✓	×	24.52	209.69	23.34	62.78	62.48	89.97
Net-3	×	×	✓	24.64	209.75	23.01	62.28	61.74	89.42
Net-4	✓	✓	×	25.04	213.19	23.48	62.99	65.71	91.13
Net-5	×	✓	✓	24.96	215.15	23.62	63.13	65.43	91.00
Net-6	✓	×	✓	25.16	213.24	23.28	62.55	62.58	89.84
Net-7	✓	✓	✓	25.48	218.65	**24.04**	**63.42**	**67.06**	**91.56**

TABLE V
FURTHER ANALYSIS OF SGF COUNTED ON TWO DATASETS

Networks	Cat	Diff	Smask	SECOND		Landsat-SCD	
				Sek	F_{scd}	Sek	F_{scd}
Net-7-1	✓	×	×	23.62	63.13	65.43	91.00
Net-7-2	×	✓	×	23.50	62.98	65.99	91.19
Net-7-3	✓	✓	×	23.77	63.28	66.21	91.26
Net-7	✓	✓	✓	**24.04**	**63.42**	**67.06**	**91.56**

TABLE VI
FURTHER ANALYSIS OF SCE COUNTED ON TWO DATASETS

Networks	label	SECOND		Landsat-SCD	
		Sek	F_{scd}	Sek	F_{scd}
Net-7	✓	24.02	63.39	**67.10**	**91.59**
Net-7	×	**24.04**	**63.42**	67.06	91.56

the best performance. By illustrating with SECOND, the enhancements achieved by Net-7 in F_{scd} are 1.23% (over Net-0), 0.80% (over Net-1), 0.64% (over Net-2), 1.14% (over Net-3), 0.43% (over Net-4), 0.29% (over Net-5), and 0.87% (over Net-6). This demonstrates that the collaboration of SGF, SCE, and MSCA significantly enhances SCGNet's ability to model key change information in dual-temporal RS images, thereby further improving the SCD results.

In addition, when we study the Param(eters) and FLOPs of different networks, we find that SGF, SCE, and MSCA are not heavy and do not significantly burden operational efficiency. For instance, when SGF is added to Net-0, the parameter volume of the model is increased by only 0.52 M, and the FLOPs are increased by 3.49 G. Similarly, after adding SCE, the growth in parameter volume and FLOPs is 0.33 M and 5.41 G, respectively. Finally, after adding MSCA, the growth in parameter volume and FLOPs is 0.44 M and 5.46 G, respectively. Compared to the performance improvements, the increases in parameters and FLOPs are well worth it.

2) Deep Study of the SGF Module: As described in Section III-B, SGF integrates two bitemporal fusion methods, channelwise concatenation and absolute difference, and also introduces SRA. Therefore, to further study the effect of SGF, it is necessary to examine the impact of each component. To this end, we built three other networks based on Net-7. Net-7-1 means that we only use $\mathbf{X}^i_{cat}$ for fusion (i.e., Net-5). Net-7-2 means that the difference feature map $\mathbf{X}^i_{diff}$ is directly used as the fusion feature. Net-7-3 means using $\mathbf{SMask}^i$ to guide the fusion of $\mathbf{X}^i_{cat}$, but not adding $\mathbf{X}^i_{diff}$. Their results are summarized in Table V, where Cat represents $\mathbf{X}^i_{cat}$, Diff represents $\mathbf{X}^i_{diff}$, and Smask represents $\mathbf{SMask}^i$. Through observation, it is found that the performance of Net-7-1 and Net-7-2 on different datasets is inconsistent, while Net-7-3 achieves better performance on all datasets than both. This shows that the knowledge contained in $\mathbf{X}^i_{cat}$ and $\mathbf{X}^i_{diff}$ is complementary to a certain extent. Among them, the former provides rich semantic information of changes, while the latter provides significant clues of differences. Despite this, the overall performance of Net-7-3 is still inferior to Net-7, which further shows that there is still redundant information in $\mathbf{X}^i_{cat}$, which may interfere with the final judgment. After introducing $\mathbf{SMask}^i$, the network can more effectively focus on areas with large semantic differences and filter out interference information unrelated to changes, thereby achieving more targeted and adaptive feature fusion and enhancing the overall effect.

3) Deep Study of the SCE Module: SCE is designed to enhance semantic consistency by injecting invariant information from the change branch into the semantic branch. To verify whether the information transferred by SCE indeed corresponds to invariant regions, we conduct an additional experiment on Net-7 by introducing label supervision to constrain the invariant information generated by the change branch explicitly. Specifically, the change probability predicted by the classifier in SCE is directly supervised using a binary cross-entropy loss with the ground-truth change labels. As shown in Table VI, the performance on both SECOND and Landsat–SCD remains almost unchanged, indicating that the network can implicitly learn invariant cues without explicit label constraints. This confirms that SCE effectively conveys invariant information to support semantic consistency modeling. To keep the architecture concise and avoid additional label requirements, no extra supervision is introduced for this component.

4) Deep Study of the MSCA Module: In practical applications, the selection of the pooling rates in MSCA has a significant impact on the final performance, which needs to be verified through experiments. To this end, we built three variants based on the Net-7 architecture by adjusting

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TABLE VII
FURTHER ANALYSIS OF MSCA COUNTED ON TWO DATASETS

Networks	Pooling rate	SECOND		Landsat-SCD	
		Sek	F_{scd}	Sek	F_{scd}
Net-7-4	$[1]$	23.61	63.12	66.28	91.30
Net-7-5	$[1, 2]$	23.47	63.22	66.80	91.47
Net-7	$[1, 2, 4]$	**24.04**	**63.42**	**67.06**	**91.56**
Net-7-6	$[1, 2, 4, 8]$	23.85	63.27	66.03	91.21

the pooling rate of MSCA: Net-7-4, Net-7-5, and Net-7-6. They use different pooling-rate combinations, and the specific results are shown in Table VII. The performance of Net-7 is also listed for comparison. From the experimental results, it can be observed that as the pooling rate gradually introduces more scales from a single scale, the performance is improved to a certain extent, especially Net-7 (using $[1, 2, 4]$), which achieves the best results on both datasets. This demonstrates that a reasonable design of the pooling rates helps our model highlight the change regions. However, when the pooling scale is further increased to $[1, 2, 4, 8]$, the performance decreases slightly. This is because the large pooling rate results in the loss of detailed information, which prevents the model from accurately identifying semantic changes. Based on the above discussion, we suggest the pooling rate in MSCA is $[1, 2, 4]$.

V. CONCLUSION

This article proposes a new SCD model, SCGNet, comprising a feature extractor, a semantic branch, a change branch, and an MSCA module. The SGF module leverages semantic relevance cues from the semantic branch to guide adaptive fusion of bitemporal features, allowing the change branch to focus more on change regions with clear semantic significance. The SCE module feeds invariant information extracted from the change branch back to the semantic branch, stabilizing bitemporal semantic representations and thus improving the overall semantic consistency of the network. In addition, the MSCA module activates and refines change features in a multiscale manner under semantic constraints, further strengthening the collaboration between semantic information and change responses. SCGNet establishes a continuous information interaction mechanism between the semantic and change branches, which is beneficial for improving semantic consistency and change localization accuracy simultaneously. The encouraging experimental results on two public datasets demonstrate the effectiveness of our SCGNet, which outperforms many existing methods.

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Xu Tang (Senior Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees in electronic circuit and system from Xidian University, Xi'an, China, in 2007, 2010, and 2017, respectively.

From 2015 to 2016, he was a Joint Ph.D. Student along with Prof. W. J. Emery at the University of Colorado at Boulder, Boulder, CO, USA. He is currently a Full Professor with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, Xidian University. His research interests include remote sensing image content-based retrieval and reranking, hyperspectral image processing, remote sensing scene classification, and object detection (for more details, please refer to: https://web.xidian.edu.cn/tangxu/en/index.html).



Weihao Jiang received the B.E. degree in artificial intelligence from Xidian University, Xi'an, China, in 2024, where he is currently pursuing the M.S. degree in computer science and technology with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education.

His research interests include remote sensing image understanding, binary change detection, and semantic change detection.



Jingjing Ma (Member, IEEE) received the B.S. and Ph.D. degrees from Xidian University, Xi'an, China, in 2004 and 2012, respectively.

She is currently an Associate Professor with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, Xidian University. Her research interests include computational intelligence and image understanding.

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Xiangrong Zhang (Senior Member, IEEE) received the B.S. and M.S. degrees from the School of Computer Science, Xidian University, Xi'an, China, in 1999 and 2003, respectively, and the Ph.D. degree from the School of Electronic Engineering, Xidian University, in 2006.

She is a Professor with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, Xidian University. From January 2015 to March 2016, she was a Visiting Scientist with the Computer Science and Artificial Intelligence Laboratory, Massachusetts Institute of Technology, Cambridge, MA, USA. Her research interests include pattern recognition, machine learning, and remote sensing image analysis and understanding.



Fang Liu (Member, IEEE) was born in China, in 1990. She received the B.S. degree in information and computing science from Henan University, Kaifeng, China, in 2012, and the Ph.D. degree in intelligent information processing from Xidian University, Xi'an, China, in 2018.

She is currently an Associate Professor at Nanjing University of Science and Technology, Nanjing, China. Her research interests include deep learning, object detection, polarimetric SAR image classification, and change detection.



Licheng Jiao (Life Fellow, IEEE) received the B.S. degree in high voltage from Shanghai Jiao Tong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees in electronic engineering from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

From 1984 to 1986, he was an Assistant Professor with the Civil Aviation Institute of China, Tianjin, China. From 1990 to 1991, he was a Post-Doctoral Fellow with the Key Laboratory for Radar Signal Processing, Xidian University, Xi'an, where he is currently the Director of the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education. He has authored or co-authored more than 200 scientific articles. His research interests include signal and image processing, nonlinear circuits and systems theory, wavelet theory, natural computation, and intelligent information processing.

Dr. Jiao is also a member of the IEEE Xian Section Executive Committee and an Executive Committee Member of Chinese Association of Artificial Intelligence. He is the Chairperson of the Awards and Recognition Committee.

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